



Javier Jareño Dorado

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Summary of CV

This section describes briefly a summary of your career in science, academic and research; the main scientific and technological achievements and goals in your line of research in the medium -and long- term. It also includes other important aspects or peculiarities.

Obtained his Master Degree in Artificial Intelligence by the Universidad Nacional de Educación a Distancia (UNED) in 2024. I currently hold the EMERGIA competitive pre-doctoral fellowship, funded by the regional Government Junta de Andalucía. Has published several journal papers, and the main research interests focus on green machine and deep learning, evolutionary algorithms, and their application to emission reduction in hybrid electric vehicles.



Javier Jareño Dorado

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Contact aut. region/reg.: Andalucía
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Current professional situation

Employing entity: Universidad de Cádiz

Department: Departamento de Ingeniería Informática, Escuela Superior de Ingeniería

Professional category: Investigador Predoctoral

Start date: 01/05/2025

Type of contract: Grant-assisted student (pre or post-doctoral, others) **Dedication regime:** Full time

Previous positions and activities

	Employing entity	Professional category	Start date
1	Universidad de Cádiz	Capítulo VI - Investigador licenciado	10/04/2024
2	Universidad de Cádiz	Capítulo VI - Investigador licenciado	03/2023
3	Universidad de Cádiz	Capítulo VI - Investigador licenciado	10/2022

1 **Employing entity:** Universidad de Cádiz **Type of entity:** University
Professional category: Capítulo VI - Investigador licenciado
Start-End date: 10/04/2024 - 30/04/2025 **Duration:** 1 year - 20 days

2 **Employing entity:** Universidad de Cádiz **Type of entity:** University
Professional category: Capítulo VI - Investigador licenciado
Start-End date: 03/2023 - 10/2023 **Duration:** 7 months

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CURRÍCULUM VITAE NORMALIZADO

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Employing entity: Universidad de Cádiz

Type of entity: University

Professional category: Capítulo VI - Investigador licenciado

Start-End date: 10/2022 - 03/2023

Duration: 6 months



Education

University education

1st and 2nd cycle studies and pre-Bologna degrees

- 1** **Name of qualification:** Máster Universitario en Investigación en Inteligencia Artificial
Degree awarding entity: Universidad Nacional de Educación a Distancia **Type of entity:** University
Date of qualification: 05/07/2024
- 2** **Name of qualification:** Graduado o Graduada en Ingeniería informática - Computación
Degree awarding entity: Universidad de Cádiz **Type of entity:** University
Date of qualification: 31/05/2022

Language skills

Language	Listening skills	Reading skills	Spoken interaction	Speaking skills	Writing skills
English	B2	B2	B2	B2	B2

Teaching experience

Experience supervising doctoral thesis and/or final year projects

- 1** **Project title:** Desarrollo de una plataforma web para gestionar competiciones de juegos de cartas y de mesa
Entity: Universidad de Cádiz **Type of entity:** University
Student: Jonas Nuñez Diaz
Obtained qualification: 10
Date of reading: 26/02/2026
- 2** **Project title:** Sistema multiplataforma para el diseño de protocolos de urgencias pediátricas
Entity: Universidad de Cádiz **Type of entity:** University
Student: Nuria Frías León
Date of reading: 19/05/2025



Courses and seminars given

- 1 **Type of event:** Seminar
Name of the event: Multiobjective optimization for the obfuscation of Software programs using LLVM
Organising entity: Universidad de Cádiz **Type of entity:** University
Hours of teaching: 6
Teaching date: 09/05/2022
Theme: Other topics
- 2 **Type of event:** Seminar
Name of the event: Intel AVX - Intrínsecos en C
Organising entity: Universidad de Cádiz **Type of entity:** University
Hours of teaching: 1
Teaching date: 10/09/2019
Theme: Other topics

Scientific and technological experience

Research and development groups/teams

Name of the group: GOAL - Graphical Methods, Optimization and Learning
Aims of the group: Design of innovative methods based on AI and graphical models, as well as their application to solve complex real world problems
Standardised code: TIC259 **Type of collaboration:** Co-authorship of projects and their development
Affiliation entity: Universidad de Cádiz **Type of entity:** University
Start date: 2022

Scientific or technological activities

R&D projects funded through competitive calls of public or private entities

- 1 **Name of the project:** Soluciones inteligentes de conducción ecológica para la movilidad sostenible (eMOB)
Entity where project took place: Universidad de Cádiz **Type of entity:** University
City of entity: Puerto Real, Andalucía, Spain
Name principal investigator (PI, Co-PI....): Patricia Ruíz Villalobos; Bernabé Dorronsoro Díaz
Nº of researchers: 6
Start-End date: 01/09/2023 - 31/08/2026
- 2 **Name of the project:** Caracterización de la eficiencia energética en el binomio software/hardware mediante análisis multifractal (eFracWare)
Entity where project took place: Universidad de Cádiz **Type of entity:** University
City of entity: Puerto Real, Andalucía, Spain



Name principal investigator (PI, Co-PI....): Paricia Ruíz Villalobos; Pablo Pavón Domínguez

Nº of researchers: 13

Start-End date: 01/12/2022 - 30/11/2024

3 Name of the project: Digitalización y valorización de la pesca en el Mediterráneo español

Degree of contribution: Researcher

Entity where project took place: Universitat de València **Type of entity:** University

City of entity: Valencia, Comunitat Valenciana, Spain

Name principal investigator (PI, Co-PI....): Paloma Herrero Racionero

Nº of researchers: 25

Type of participation: Team member

Start-End date: 01/10/2023 - 31/10/2023

Total amount: 486.335,86 €

Dedication regime: Full time

4 Name of the project: INTELLIGENT GENERATION OF SUSTAINABLE SOFTWARE - GENIUS (P18-FR-2399)

Degree of contribution: Researcher

Entity where project took place: Universidad de Cádiz **Type of entity:** University

City of entity: Puerto Real, Andalucía, Spain

Name principal investigator (PI, Co-PI....): Bernabé Dorronsoro Profesor Titular de Universidad

Nº of researchers: 15

Type of participation: Team member

Start-End date: 01/01/2020 - 31/03/2023

Total amount: 120.000 €

Dedication regime: Full time

Scientific and technological activities

Scientific production

Publications, scientific and technical documents

1 Javier Jareño; José Miguel Aragón Jurado; Juan Carlos de la Torre; Patricia Ruiz; Bernabé Dorronsoro. Energy-efficient large language models. Future Generation Computer Systems. Elsevier, 19/03/2026.

Type of production: Scientific paper

Format: Journal

Total no. authors: 5

Corresponding author: Yes

2 Juan Carlos de la Torre Macías; Javier Jareño Dorado; José Miguel Aragón Jurado; Sébastien Varette; Bernabé Dorronsoro Díaz. Source Code Obfuscation with Genetic Algorithms Using LLVM Code Optimizations. Logic Journal of the IGPL. Oxford Academic, 14/05/2024.

Type of production: Scientific paper

Format: Book

Corresponding author: No

3 Javier Jareño Dorado; Guillermo Bárcena González; Jairo Castro Gutiérrez; Remedios Cabrera Castro; Pedro Luís Galindo Riaño. Enhancing Fish Auction with Deep Learning and Computer Vision: Automated Caliber and Species Classification. Fishes. 9 - 4, MDPI, 13/04/2024.



Type of production: Scientific paper

Format: Journal

Corresponding author: No

- 4** Javier Jareño Dorado; Guillermo Bárcena González; Jairo Castro Gutiérrez; Remedios Cabrera Castro; Pedro Luis Galindo Riaño. Automatic labeling of fish species using deep learning across different classification strategies. Frontiers in Computer Science. 6, Frontiers, 29/02/2024.

Type of production: Scientific paper

Format: Journal

Corresponding author: No

- 5** Jose Miguel Aragón Jurado; Juan Carlos de la Torre Macías; Javier Jareño Dorado; Bernabé Dorronsoro Díaz; Albert Zomaya; Patricia Ruíz Villalobos. Neuroevolved bi-directional LSTM applied to zero emission zones management in urban transport. Applied Soft Computing. 148, pp. 110943. Elsevier, 01/11/2023.

Type of production: Scientific paper

Format: Journal

- 6** Bernabé Dorronsoro Díaz; José Miguel Aragón Jurado; Javier Jareño Dorado; Juan Carlos de la Torre Macías; Patricia Ruíz Villalobos. A Survey on Automatic Source Code Transformation for Green Software Generation. Martin A. Abraham, 2024.

Type of production: Bibliographic review

Format: Journal

Works presented at national or international conferences

- 1** **Title of the work:** Problema de Optimización de Carga Dinámica
Name of the conference: Doctoral Consortium on Computer Science (JIPII 2026)

Corresponding author: Yes

City of event: Cadiz, Andalucía, Spain

Date of event: 22/06/2026

End date: 22/06/2026

Organising entity: Universidad de Cádiz

Type of entity: University

City organizing entity: Cadiz, Andalucía, Spain

Javier Jareño; Cintia Córcoles Cabeza; Patricia Ruiz; Bernabé Dorronsoro. "Problema de Optimización de Carga Dinámica".

- 2** **Title of the work:** Analyzing the Impact of Travel Time on the Driving Assignment Strategies of Heavy Duty Trucks

Name of the conference: Transport Research Arena, TRA 2026

Corresponding author: Yes

City of event: Budapest, Hungary

Date of event: 18/05/2026

End date: 21/05/2026

Organising entity: Comisión Europea

Type of entity: European

City organizing entity: Budapest, Hungary

Javier Jareño Dorado; Bernabé Dorronsoro Díaz; Patricia Ruíz Villalobos. "Analyzing the Impact of Travel Time on the Driving Assignment Strategies of Heavy Duty Trucks".

- 3** **Title of the work:** Dynamic Load Optimization Problem
Name of the conference: International Conference on Optimization and Learning (OLA2026)

Corresponding author: Yes

City of event: Chania, Greece

Date of event: 28/04/2026

End date: 30/04/2026

Organising entity: University of Lille

Type of entity: University



City organizing entity: Lille, France

Javier Jareño Dorado; Patricia Ruiz; Bernabé Dorronsoro Díaz. "Dynamic Load Optimization Problem".

4 Title of the work: GreenLLMs: Sustainable Large Language Models

Name of the conference: International Conference on Optimization and Learning (OLA2026)

Corresponding author: Yes

City of event: Chania, Greece

Date of event: 28/04/2026

End date: 30/04/2026

Organising entity: University of Lille

Type of entity: University

City organizing entity: Lille, France

Javier Jareño Dorado; José Miguel Aragón Jurado; Juan Carlos de la torre Macías; Patricia Ruiz; Bernabé Dorronsoro Díaz. "GreenLLMs: Sustainable Large Language Models".

5 Title of the work: A Two-step Approach to Find Short Compilation Transformation Sequences for Optimal Software Runtime Performance

Name of the conference: Genetic and Evolutionary Computation Conference (GECCO)

City of event: Málaga, Andalucía, Spain

Date of event: 14/07/2025

End date: 18/07/2025

Organising entity: Association for Computing Machinery

Type of entity: Associations and Groups

Juan Carlos de la Torre; José Miguel Aragón Jurado; Javier Jareño; Bernabé Dorronsoro; Patricia Ruiz. "A Two-step Approach to Find Short Compilation Transformation Sequences for Optimal Software Runtime Performance".

6 Title of the work: Optimización en dos pasos del rendimiento del software usando transformaciones de compilación

Name of the conference: Metaheurísticas, Algoritmos Evolutivos y Bioinspirados (MAEB)

Corresponding author: No

City of event: Donostia/San Sebastian, País Vasco, Spain

Date of event: 28/05/2025

End date: 30/05/2025

Organising entity: Universidad del País Vasco

Type of entity: University

City organizing entity: Donostia,

Juan Carlos de la Torre; José Miguel Aragón; Javier Jareño; Bernabé Dorronsoro; Patricia Ruiz.

"Optimización en dos pasos del rendimiento del software usando transformaciones de compilación".

7 Title of the work: Two-Level Software Obfuscation with Cooperative Co-Evolutionary Algorithms

Name of the conference: 2024 IEEE Congress on Evolutionary Computation, CEC 2024

Corresponding author: No

City of event: Yokohama, Japan

Date of event: 30/09/2024

End date: 05/07/2024

Organising entity: IEEE

José Miguel Aragón Jurado; Javier Jareño Dorado; Juan Carlos de la Torre Macías; Patricia Ruíz Villalobos; Bernabé Dorronsoro Díaz. "Two-Level Software Obfuscation with Cooperative Co-Evolutionary Algorithms".

8 Title of the work: Ofuscación de Software en dos Niveles usando Algoritmos Cooperativos Coevolutivos

Name of the conference: XX Conferencia de la Asociación Española para la Inteligencia Artificial

Corresponding author: No

City of event: A Coruña, Galicia, Spain



Date of event: 19/06/2024

End date: 21/06/2024

Organising entity: Universidad de La Coruña

Type of entity: University

City organizing entity: A Coruña, Galicia, Spain

José Miguel Aragón; Javier Jareño; Juan Carlos de la Torre; Patricia Ruiz; Bernabé Dorronsoro.

"Ofuscación de Software en dos Niveles usando Algoritmos Cooperativos Coevolutivos".

- 9 Title of the work:** Management of Zero Emissions Zones in Urban Transport by Means of Neuroevolution
Name of the conference: Tenth Spanish-German Symposium on Applied Computer Science (SGSOACS 2024)

Corresponding author: Yes

City of event: Cadiz, Andalucía, Spain

Date of event: 12/06/2024

End date: 14/06/2024

Organising entity: Universidad de Cádiz

Type of entity: University

City organizing entity: Cadiz, Andalucía, Spain

José Miguel Aragón Jurado; Javier Jareño; Bernabé Dorronsoro; Patricia Ruiz. "Management of Zero Emissions Zones in Urban Transport by Means of Neuroevolution".

- 10 Title of the work:** Multifractal Analysis of Thermal Images of Electronic Devices in Different Colour Profiles
Name of the conference: ADVCOMP 2023 : The Seventeenth International Conference on Advanced Engineering Computing and Applications in Sciences

Corresponding author: No

City of event: Porto, Portugal

Date of event: 25/09/2023

End date: 29/09/2023

Organising entity: International Academy, Research, and Industry Association

Type of entity: University Research Institute

City organizing entity: Porto, Portugal

Marina Díaz Jiménez; Juan Carlos de la Torre Macías; Javier Jareño Dorado; Patricia Ruíz Villalobos; Bernabé Dorronsoro Díaz; Pablo Pavón Domínguez. "Multifractal Analysis of Thermal Images of Electronic Devices in Different Colour Profiles".

- 11 Title of the work:** Characterization and categorization of software programs on x86 architectures
Name of the conference: INTERNATIONAL CONFERENCE ON OPTIMIZATION AND LEARNING (OLA2023)

Corresponding author: No

City of event: Málaga, Andalucía, Spain

Date of event: 03/05/2023

End date: 05/05/2023

Organising entity: Universidad de Málaga

Type of entity: University

City organizing entity: Málaga, Andalucía, Spain

Javier Jareño Dorado; Juan Carlos de la torre Macías; Bernabé Dorronsoro Díaz. "Characterization and categorization of software programs on x86 architectures".

- 12 Title of the work:** Sistemas de caracterización y agrupación de benchmarks en arquitecturas x86
Name of the conference: Congreso SISTEDES - Jornadas de Ingeniería del Software y Bases de Datos (JISBD)

Type of event: Conference

City of event: Santiago de Compostela, Galicia, Spain

Date of event: 05/09/2022

End date: 07/09/2022

**Organising entity:** COGRADE**Type of entity:** Public Research Body**City organizing entity:** Santiago de Compostela, Galicia, Spain

Javier Jareño Dorado; Juan Carlos de la Torre Macías; Bernabé Dorronsoro Díaz. "International Conference on Optimization and Learning". 1824, pp. 327 - 340. ISBN 978-3-031-34019-2

- 13 Title of the work:** Obfuscating LLVM Intermediate Representation Source Code with NSGA-II
Name of the conference: International Conference on Computational Intelligence in Security for Information Systems. CISIS 2022
City of event: Salamanca, Castilla y León, Spain
Date of event: 05/09/2022
End date: 07/09/2022
Organising entity: Grupo de investigación BISITE **Type of entity:** Public Research Body
City organizing entity: Salamanca, Castilla y León, Spain
 Juan Carlos de la Torre Macías; Jose Miguel Aragón Jurado; Javier Jareño Dorado; Sebastien Varette; Bernabé Dorronsoro Díaz. "15th Intl. Conf. on Computational Intelligence in Security for Information Systems (CISIS'22)".

Other achievements

Obtained grants and scholarships

- 1 Name of the grant:** Estrategia de Investigación, Desarrollo e Innovación de Andalucía (EIDIA)
Aims: Pre-doctoral
Awarding entity: Junta de Andalucía **Type of entity:** Regional
Conferral date: 01/06/2025 **Duration:** 5 months
End date: 30/11/2025
Entity where activity was carried out: Universidad de Cádiz
Faculty, institute or centre: Escuela Superior de Ingeniería
- 2 Name of the grant:** Formación de Profesorado Universitario 2024
Aims: Pre-doctoral
Awarding entity: Ministerio de Ciencia, Innovación y Universidades **Type of entity:** National
Conferral date: 12/11/2025
Entity where activity was carried out: Universidad de Cádiz
Faculty, institute or centre: Escuela Superior de Ingeniería
- 3 Name of the grant:** Beca de Colaboración MEFP 2021/2022
Aims: Bachelor Grant
Awarding entity: Ministerio de Educación, Formación Profesional y Deportes **Type of entity:** State agency
- 4 Name of the grant:** INICIA-INV: Iniciación a la investigación
Aims: Bachelor Grant
Awarding entity: Universidad de Cádiz **Type of entity:** University



Prizes, mentions and distinctions

Description: Premio extraordinario Fin de Carrera

Awarding entity: Universidad de Cádiz

Type of entity: University

City awarding entity: Puerto Real, Andalucía, Spain

Conferral date: 03/02/2023

Other distinctions (professional or business career)

Description: Ganador Olimpiadas de Física Provincia de Cádiz

Awarding entity: Universidad de Cádiz

Type of entity: University

City awarding entity: Puerto Real, Andalucía, Spain

Conferral date: 21/03/2018